Physical Chemistry (II)

Lecture 12

Zeeman Effect (2)
Perturbation Theory

Where Are We?

기본 원리	응용
단일 입자 계	PIB, SHO, RR 수소 원자
다입자 계	나머지 모든 원자 모든 분자

Where Are We: Hydrogen Atom

$\hat{H}_0$과 고유함수 공유

$\hat{H}_0$과 고유함수 공유 X

$\hat{H}_0$

$\hat{H}_0 + \hat{H}_{\mathrm{magnet}}$

$\hat{H}_0 + \underbrace{\hat{H}_{\mathrm{rel}} + \hat{H}_{\mathrm{SO}}}$

"그냥" 수소 원자 → 자기장 속 H → 미세 구조

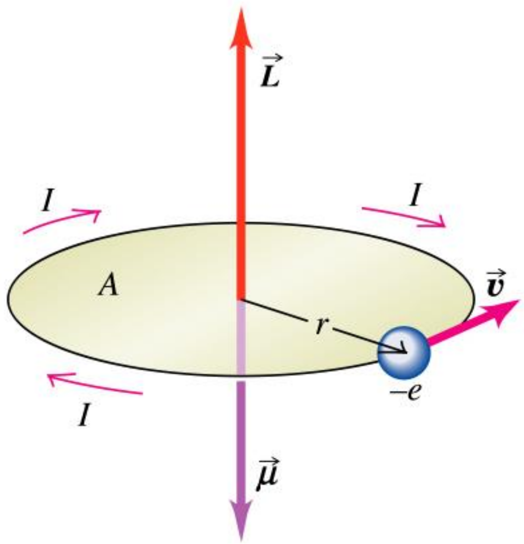
- 오비탈

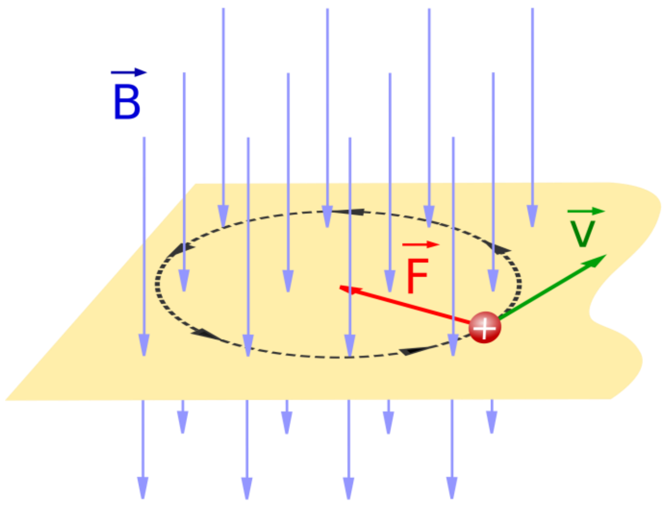
- 스핀
- 각운동량의 합

- 항 기호

섭동론

Electron as a Magnet





$$\vec{\mu} = -\frac{e}{2}(\vec{r} \times \vec{v}) = -\frac{e}{2m}\vec{L}$$

$$V = \frac{eB}{2m}L_z$$

(Image: https://dpotoyan.github.io/Chem324/Lec5-2.html)

Last class

Quantum Mechanics

$$\left(-\frac{\hbar^2}{2\mu}\nabla^2 + \mathcal{V}\right)\psi = E\psi$$

If a hydrogen-atom electron is in a magnetic field,

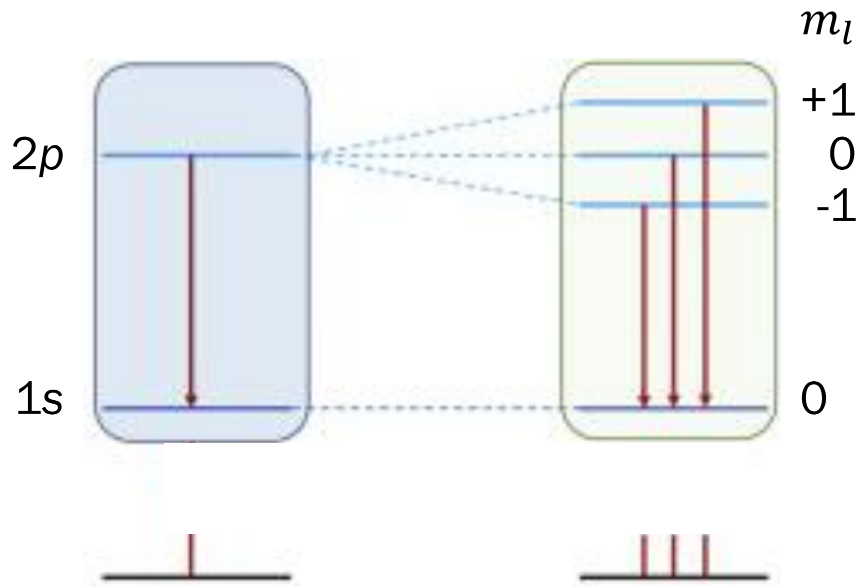
$$\left(\underbrace{-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}}_{\substack{\text{Hamiltonian for H atom} \\ \hat{H}_0}} + \underbrace{\frac{eB}{2\mu}\hat{L}_z}_{\substack{\text{Additional Hamiltonian} \\ \hat{H}_{\text{magnet}}}}\right)\psi = E\psi$$

The eigenfunctions for $\hat{H}_0$ are also those for $\hat{H}_{\text{magnet}}$

Last class

Effect of Magnetic Field

$$E = -\frac{ZR_{\mathrm{H}}}{n^2} + \beta_B m_l B$$



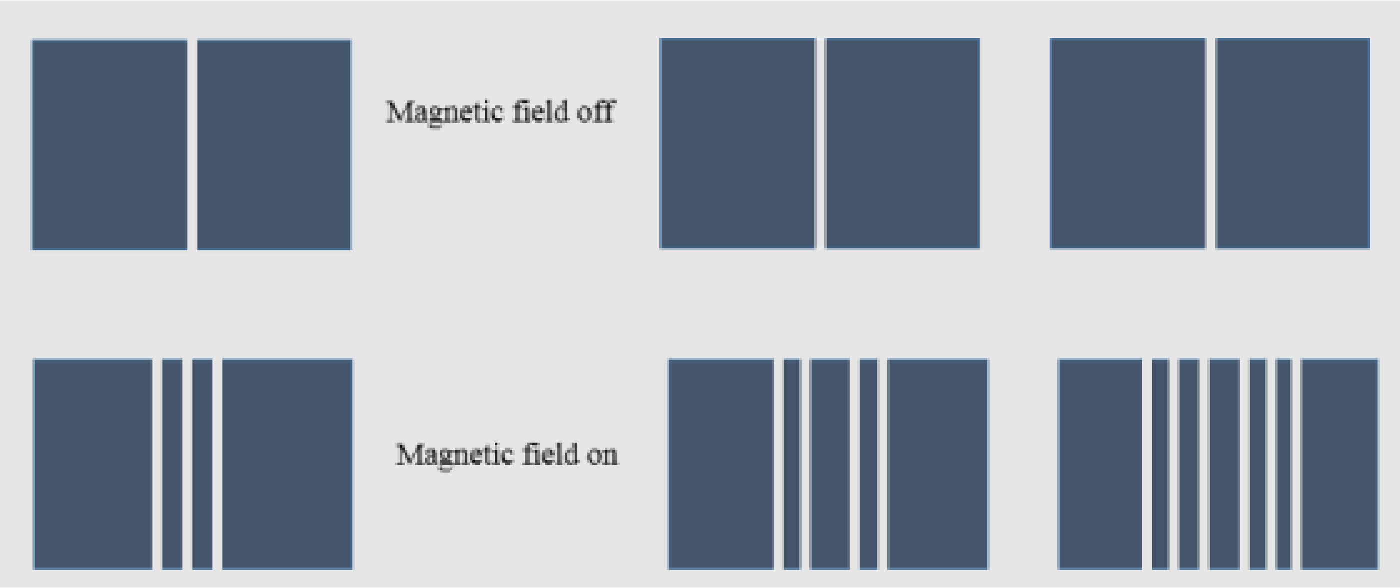
(Image: https://www.sciencedirect.com/science/article/pii/B9780444642257000031)

Last class

Zeeman Effect

Normal Zeeman effect

Anomalous Zeeman effect



(Image: https://www.differencebetween.com/difference-between-normal-and-vs-anomalous-zeeman-effect/)

Last class

"Spin" Angular Momentum

An electron has *intrinsic* angular momentum

= an electron intrinsically behaves like a tiny magnet

- Eigenfunction $|s\ m_s\rangle$

$$\hat{S}^2|s\ m_s\rangle = s(s+1)\hbar^2|s\ m_s\rangle$$

$$\hat{S}_z|s\ m_s\rangle = m_s\hbar|s\ m_s\rangle$$

- Quantum numbers s, m_s

s: fixed (intrinsic)

$$m_s = -s, -(s-1), \cdots, (s-1),$$

Last class

Free Electron in Magnetic Field

- For an electron, $s = 1/2$

$$m_s = \pm 1/2$$

- Orbital angular momentum energy

$$E_{\text{orbital}} = \beta_{\text{B}} m_l B$$

- Spin angular momentum energy

$$E_{\text{spin}} = g\beta_{\text{B}} m_s B$$

- $g \approx 2$ (g-factor): experimentally required

Last class

Full Description of H-Atom Electron

- Quantum numbers: n, l, m_l, m_s (s is fixed)
- Wavefunction:

$$\psi_{nlm_l}(r,\theta,\phi)|m_s\rangle = |n\ l\ m_l\ m_s\rangle$$

- Energy (no magnetic field):

$$E_n = -\frac{ZR_{\mathrm{H}}}{n^2}$$

- Energy (in a magnetic field):

$$E = -\frac{ZR_{\mathrm{H}}}{n^2} + ?$$

RS 8.6
M 7.8
* Griffiths 7.4

Orbital + Spin under $\vec{B}$

$$\vec{\mu}_{\text{orbit}} = -\frac{e}{2m}\vec{L}, \qquad \vec{\mu}_{\text{spin}} = -\frac{eg}{2m}\vec{S} \approx -\frac{e}{m}\vec{S}$$

From the potential energy $V = -\vec{\mu}\cdot\vec{B}$,

$$V = -\left(-\frac{e}{2m}\vec{L} - \frac{e}{m}\vec{S}\right)\cdot\vec{B} = \frac{e}{2m}\left(\vec{L}+2\vec{S}\right)\cdot\vec{B}$$

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{e}{2\mu}\left(\vec{L}+2\vec{S}\right)\cdot\vec{B}$$

external field + induced field

In a Strong Field

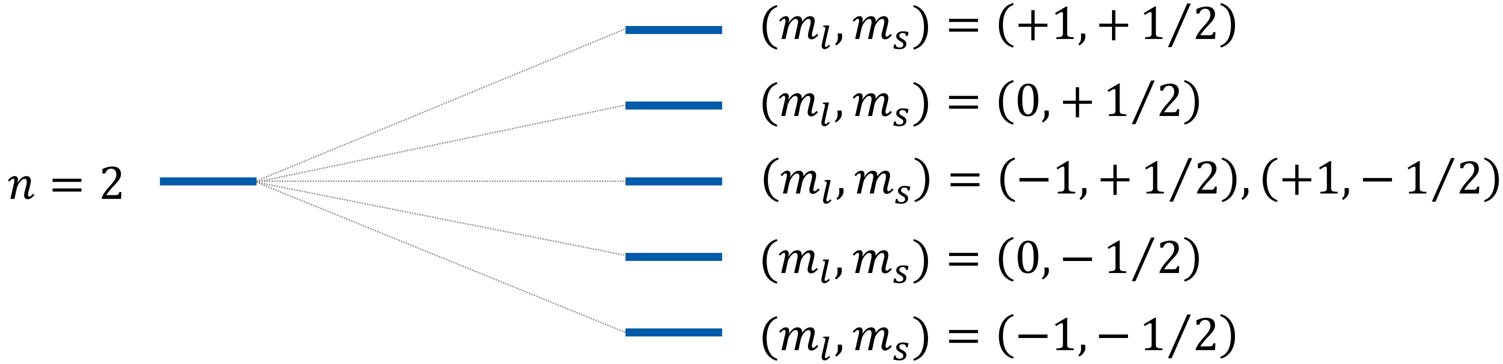
$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{e}{2\mu}\left(\vec{L} + 2\vec{S}\right)\cdot\vec{B}$$

Strong field: $\vec{B} = \vec{B}_{\mathrm{ext}} = B\hat{k}$

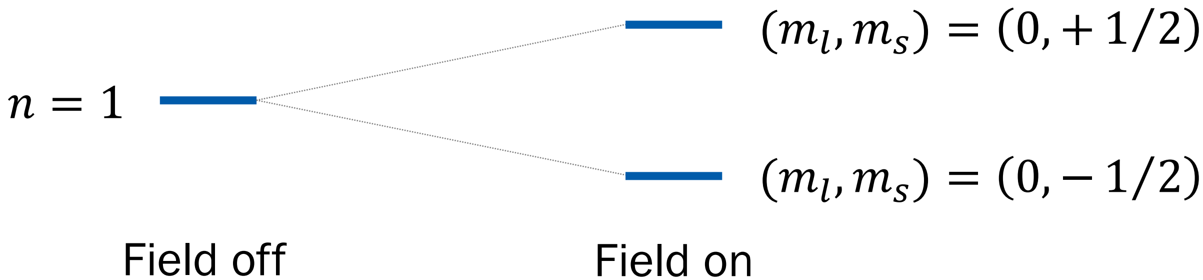
$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{eB}{2\mu}\left(\hat{L}_z + 2\hat{S}_z\right)$$

$$E = E_n + E_{\mathrm{magnet}} = -Z\frac{R_H}{n^2} + \beta_{\mathrm{B}}B(m_l + 2m_s)$$

Strong-Field Energy Levels

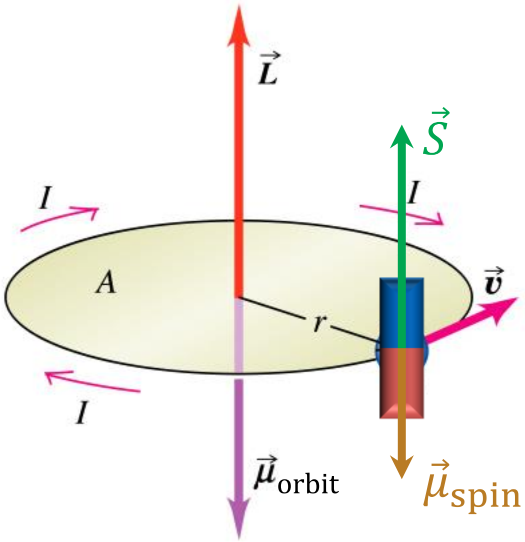


$$\boxed{E = -Z\frac{R_H}{n^2} + \beta_\mathrm{B} B(m_l + 2m_s)}$$



In a Weak Field

Need to consider the spin-orbit coupling

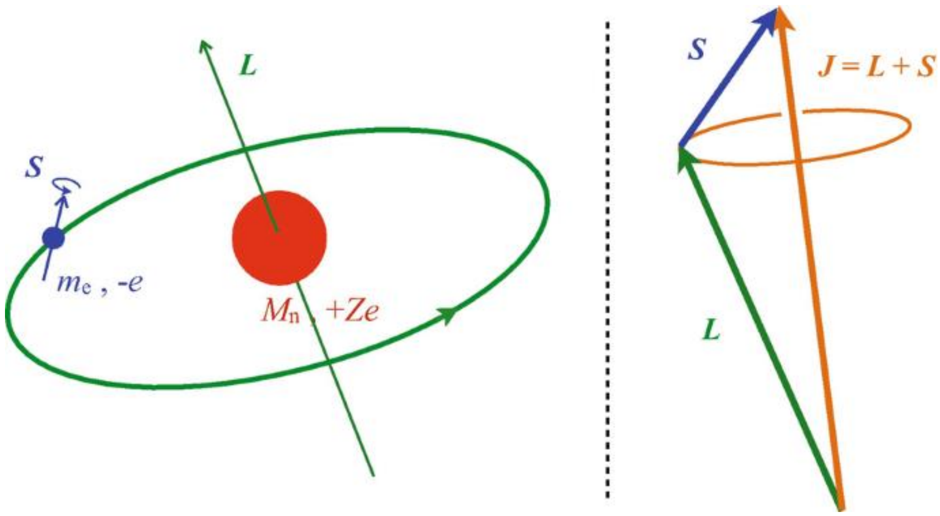


How?

(Image: https://dpotoyan.github.io/Chem324/Lec5-2.html)

Total Angular Momentum

- Orbital and spin angular momenta can be added
- Angular momenta are vectors ($\vec{L}$ and $\vec{S}$)
- The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ is conserved



(Image: Kastberg, "The Spin–Orbit Interaction", in *Structure of Multielectron Atoms*, 2020)

M 7.6

QM of Total Angular Momentum

- Eigenfunctions $|j\ m_j\rangle$
- Eigenvalues

$$\hat{J}^2|j\ m_j\rangle = j(j+1)\hbar^2|j\ m_j\rangle$$

$$\hat{J}_z|j\ m_j\rangle = m_j\hbar|j\ m_j\rangle$$

- Quantum numbers
  - Allowed $m_j = -j, -(j-1), \cdots, (j-1), j$
  - Allowed $j = (l+s), (l+s-1), \cdots, |l-s|$

Examples

Allowed $j = (l+s), (l+s-1), \cdots, |l-s|$

- $l = 0$ (s orbitals): $j = 1/2$
  - $j = 1/2 \longrightarrow m_j = \pm 1/2$
- $l = 1$ (*p* orbitals): $j = 3/2\,, 1/2$
  - $j = 3/2 \longrightarrow m_j = \pm 3/2\,, \pm 1/2$
  - $j = 1/2 \longrightarrow m_j = \pm 1/2$
- $l = 2$ (*d* orbitals): $j = 5/2\,, 3/2$
  - $j = 5/2 \longrightarrow m_j = \pm 5/2\,, \pm 3/2\,, \pm 1/2$
  - $j = 3/2 \longrightarrow m_j = \pm 3/2\,, \pm 1/2$

Two Descriptions of States

	m_l and m_s	j and m_j
1s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
2s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
2p ($l = 1$)	(+1, +1/2), (+1, -1/2) (0, +1/2), (0, -1/2) (-1, +1/2), (-1, -1/2)	(3/2, +3/2), (3/2, -3/2) (3/2, +1/2), (3/2, -1/2) (1/2, +1/2), (1/2, -1/2)
3s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
3p ($l = 1$)	(+1, +1/2), (+1, -1/2) (0, +1/2), (0, -1/2) (-1, +1/2), (-1, -1/2)	(3/2, +3/2), (3/2, -3/2) (3/2, +1/2), (3/2, -1/2) (1/2, +1/2), (1/2, -1/2)
3d ($l = 2$)	10 states	10 states

Not in text

Weak-Field Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{e}{2\mu}\left(\vec{L} + 2\vec{S}\right) \cdot \vec{B}$$

Here, $\vec{L} + 2\vec{S} = \vec{J} + \vec{S}$ and on average,

$$\vec{S}_{\text{avg}} = \vec{S} \cdot \left(\frac{\vec{J}}{|\vec{J}|}\right)\frac{\vec{J}}{|\vec{J}|}$$

$$\vec{L} + 2\vec{S} \approx \vec{J} + \vec{S}_{\text{avg}} = \vec{J}\left(1 + \frac{\vec{S} \cdot \vec{J}}{J^2}\right)$$

Not in text

Vector Calculation

$$\vec{L} + 2\vec{S} \approx \vec{J} + \vec{S}_{\text{avg}} = \vec{J}\left(1 + \frac{\vec{S} \cdot \vec{J}}{J^2}\right)$$

Since $\vec{L} = \vec{J} - \vec{S}$, $L^2 = J^2 + S^2 - 2\vec{S} \cdot \vec{J}$ and

$$\vec{S} \cdot \vec{J} = \frac{1}{2}(J^2 + S^2 - L^2)$$

Now,

$$\vec{L} + 2\vec{S} \approx \vec{J}\left[1 + \frac{1}{2J^2}(J^2 + S^2 - L^2)\right]$$

Weak-Field Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{e}{2\mu}\left[1 + \frac{1}{2\hat{J}^2}\left(\hat{J}^2 + \hat{S}^2 - \hat{L}^2\right)\right]\vec{J}\cdot\vec{B}$$

If we choose the z-axis to lie along $\vec{B}$,

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{eB}{2\mu}\left[1 + \frac{1}{2\hat{J}^2}\left(\hat{J}^2 + \hat{S}^2 - \hat{L}^2\right)\right]\hat{J}_z$$

$$E = -Z\frac{R_H}{n^2} + \beta_{\mathrm{B}}B\underbrace{\left(1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}\right)}_{\text{Landé g factor } g(j,l)}m_j$$

g Factor Calculation

$$g(j,l) = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

For $l = 1$,

- $j = 3/2$:

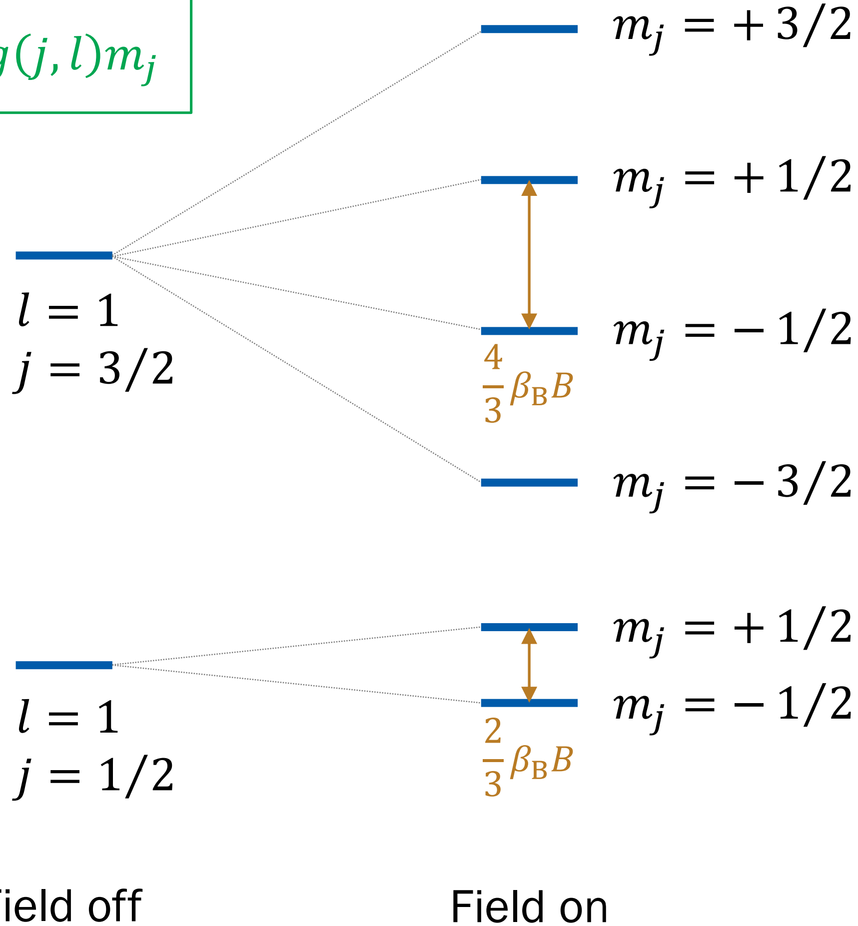
$$g(j,l) = 1 + \frac{(3/2)(3/2+1)+(1/2)(1/2+1)-1(1+1)}{2\cdot(3/2)(3/2+1)} = 1 + \frac{1}{3} = \frac{4}{3}$$

- $j = 1/2$:

$$g(j,l) = 1 + \frac{(1/2)(1/2+1)+(1/2)(1/2+1)-1(1+1)}{2\cdot(1/2)(1/2+1)} = 1 - \frac{1}{3} = \frac{2}{3}$$

Weak-Field Energy Levels

$$E = -Z\frac{R_H}{n^2} + \beta_{\mathrm{B}} B g(j,l) m_j$$



Not in text

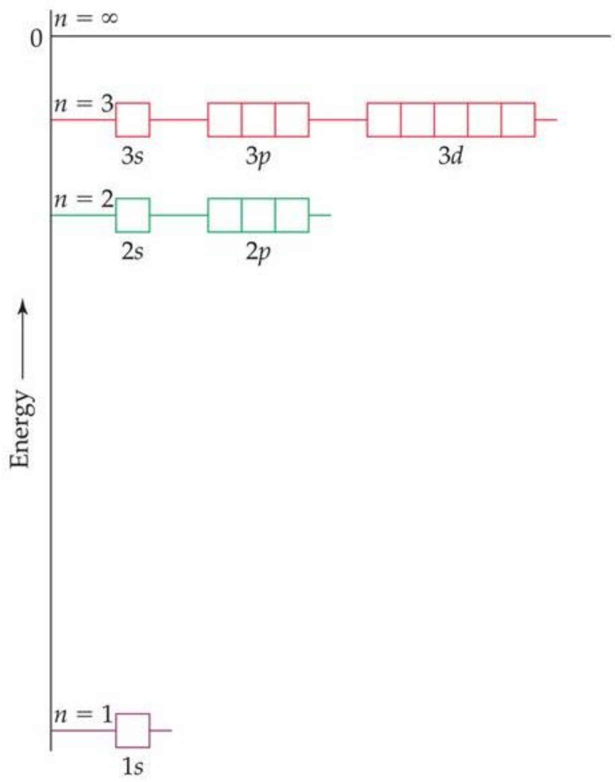
"Good" Quantum Numbers

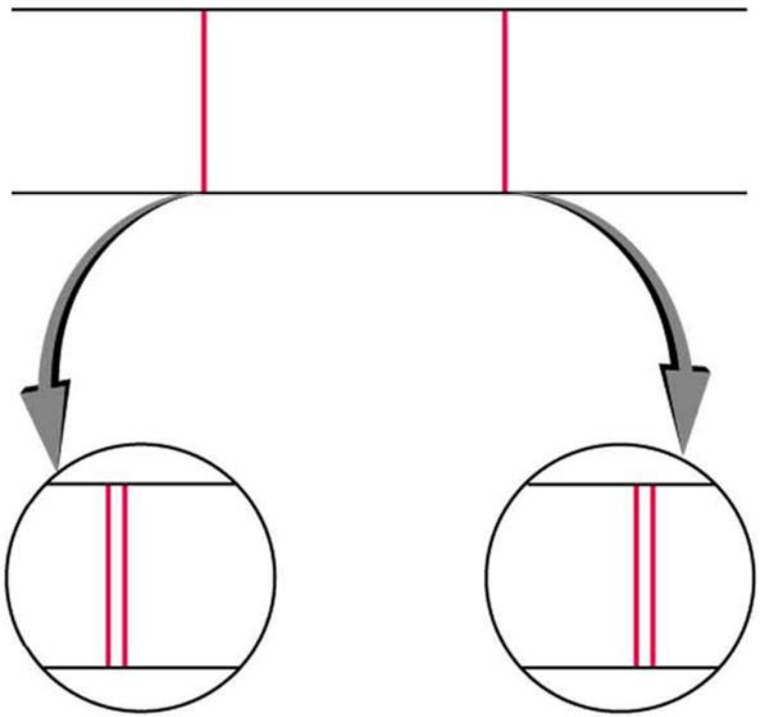
Two sets of quantum numbers: (n, l, m_l, m_s) and (n, l, j, m_j)

- Strong $\vec{B}$: (n, l, m_l, m_s) describes the state better
- Weak $\vec{B}$: (n, l, j, m_j) describes the state better

M 7.7

Fine Structure of Hydrogen





(Image: https://slideplayer.com/slide/5959399/
http://pressbooks-dev.oer.hawaii.edu/collegephysics/chapter/30-7-patterns-in-spectra-reveal-more-quantization/)

Origin of Fine Structure

1. Relativistic effect
2. Spin-orbit coupling

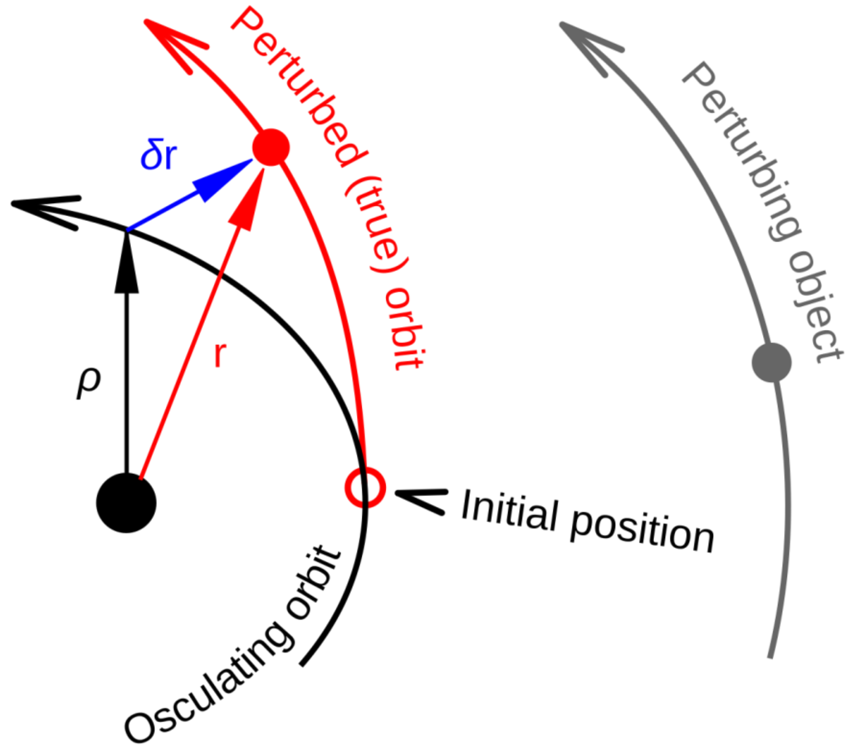
$$\hat{H}_0 \rightarrow \hat{H}_0 + \hat{H}_{\mathrm{rel}} + \hat{H}_{\mathrm{so}}$$

$\hat{H}_{\mathrm{rel}}$ or $\hat{H}_{\mathrm{so}}$ do NOT share eigenfunctions with $\hat{H}_0$!

(Note: the two effects are relatively small)

RS 5.4
M 8.5
* Griffiths 7.1

“Perturbation”



(Image: https://commons.wikimedia.org/wiki/File:Enckes_method-vector.svg)

Perturbation Theory

$$\hat{H}\psi = E\psi$$

We can apply the perturbation theory if the Hamiltonian can be decomposed as

unperturbed Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

perturbation

where we know $E_n^{(0)}$ and $\psi_n^{(0)}$ (n: quantum number):

$$\hat{H}_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

- Note: $\hat{H}_0$ and $\hat{H}'$ do not share the eigenfunctions

Series Expansion

Introduce the bookkeeping parameter, λ:

$$\hat{H} = \hat{H}_0 + \lambda \hat{H}'$$

and assume that

$$\psi_n = \psi_n^{(0)} + \lambda \psi_n^{(1)} + \lambda^2 \psi_n^{(2)} + \cdots$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots$$

Now substitute everything into

$$\hat{H} \psi_n = E_n \psi_n$$

Collecting Terms

$$\left(\hat{H}_0 + \lambda\hat{H}'\right)\left[\psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \cdots\right]$$
$$= \left[E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots\right]\left[\psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \cdots\right]$$

Collecting like powers of λ,

$$\hat{H}_0\psi_n^{(0)} + \lambda\left[\hat{H}_0\psi_n^{(1)} + \hat{H}'\psi_n^{(0)}\right] + \lambda^2\left[\hat{H}_0\psi_n^{(2)} + \hat{H}'\psi_n^{(1)}\right] + \cdots$$
$$= E_n^{(0)}\psi_n^{(0)} + \lambda\left[E_n^{(0)}\psi_n^{(1)} + E_n^{(1)}\psi_n^{(0)}\right]$$
$$+ \lambda^2\left[E_n^{(0)}\psi_n^{(2)} + E_n^{(1)}\psi_n^{(1)} + E_n^{(2)}\psi_n^{(0)}\right] + \cdots$$

RS 5.4
M 8.5

Different Orders

Zeroth order (λ^0):

$$\hat{H}_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)}$$ (what we knew)

First order (λ^1):

$$\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

Second order (λ^2):

$$\hat{H}_0 \psi_n^{(2)} + \hat{H}' \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

$$\vdots$$

RS 5.4
M 8.5

First-Order Correction

$$\hat{H}_0\psi_n^{(1)} + \hat{H}'\psi_n^{(0)} = E_n^{(0)}\psi_n^{(1)} + E_n^{(1)}\psi_n^{(0)}$$

Multiply both sides by $\psi_n^{(0)*}$ and integrate;

$$\int \psi_n^{(0)*}\hat{H}_0\psi_n^{(1)}\,\mathrm{d}\tau + \int \psi_n^{(0)*}\hat{H}'\psi_n^{(0)}\,\mathrm{d}\tau$$

$$= E_n^{(0)}\int \psi_n^{(0)*}\psi_n^{(1)}\,\mathrm{d}\tau + E_n^{(1)}\underbrace{\int \psi_n^{(0)*}\psi_n^{(0)}\,\mathrm{d}\tau}_{= 1 \text{ (normalized)}}$$

RS 5.4
M 8.5

Some Math

$$\int \psi_n^{(0)*} \widehat{H}_0 \psi_n^{(1)}\, \mathrm{d}\tau + \int \psi_n^{(0)*} \widehat{H}' \psi_n^{(0)}\, \mathrm{d}\tau = E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)}\, \mathrm{d}\tau + E_n^{(1)}$$

For a QM (Hermitian) operator $\hat{A}$, we have

$$\int f^*\left(\hat{A}g\right) \mathrm{d}\tau = \int \left(\hat{A}f\right)^* g\, \mathrm{d}\tau$$

Hence, the first term in LHS becomes

$$\int \psi_n^{(0)*} \widehat{H}_0 \psi_n^{(1)}\, \mathrm{d}\tau = \int \left[\widehat{H}_0 \psi_n^{(0)}\right]^* \psi_n^{(1)}\, \mathrm{d}\tau = E_n^{(0)*} \int \psi_n^{(0)*} \psi_n^{(1)}\, \mathrm{d}\tau$$

which cancels the first term in RHS.

RS 5.4
M 8.5

Finally

$$\cancel{\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)}\, \mathrm{d}\tau} + \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)}\, \mathrm{d}\tau = \cancel{E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)}\, \mathrm{d}\tau} + E_n^{(1)}$$

Therefore,

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)}\, \mathrm{d}\tau$$

One can also obtain (just for your information)

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\int \psi_m^{(0)*} \hat{H}' \psi_n^{(0)}\, \mathrm{d}\tau}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

Example 8-5

Calculate the first-order correction to the ground-state energy of an anharmonic oscillator whose potential is

$$V(x) = \frac{1}{2}kx^2 + \frac{1}{6}\gamma_3 x^3 + \frac{1}{24}\gamma_4 x^4.$$

The perturbation is

$$\hat{H}' = \frac{1}{6}\gamma_3 x^3 + \frac{1}{24}\gamma_4 x^4$$

and the ground-state wavefunction of a SHO is

$$\psi_0(x) = (\alpha/\pi)^{1/4} e^{-\alpha x^2/2} \text{ (where } \alpha = (k\mu)^{1/2}/\hbar\text{)}$$

Hence,

$$E_0^{(1)} = \int_{-\infty}^{\infty} \psi_0^{(0)*} \hat{H}' \psi_0^{(0)} \,\mathrm{d}x = \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} e^{-\alpha x^2/2} \left(\frac{1}{6}\gamma_3 x^3 + \frac{1}{24}\gamma_4 x^4\right) e^{-\alpha x^2/2} \,\mathrm{d}x$$

Example 8-5

Calculate the first-order correction to the ground-state energy of an anharmonic oscillator whose potential is

$$V(x) = \frac{1}{2}kx^2 + \frac{1}{6}\gamma_3 x^3 + \frac{1}{24}\gamma_4 x^4.$$

$$E_0^{(1)} = \left(\frac{\alpha}{\pi}\right)^{1/2}\left(\frac{\gamma_3}{6}\int_{-\infty}^{\infty} x^3 e^{-\alpha x^2}\,\mathrm{d}x + \frac{\gamma_4}{24}\int_{-\infty}^{\infty} x^4 e^{-\alpha x^2}\,\mathrm{d}x\right)$$

The first term is zero, as the integrand is an odd function. For the second term, we use the table of integrals to obtain

$$E_0^{(1)} = \left(\frac{\alpha}{\pi}\right)^{1/2} \times \frac{\gamma_4}{24} \times \frac{3\pi^{1/2}}{4\alpha^{5/2}} = \frac{\gamma_4}{32\alpha^2}$$

Where Are We: Hydrogen Atom

$\hat{H}_0$과 고유함수 공유

$\hat{H}_0$과 고유함수 공유 X

$\hat{H}_0$

"그냥" 수소 원자

- 오비탈

$\hat{H}_0 + \hat{H}_{\mathrm{magnet}}$

자기장 속 H

- 스핀
- 각운동량의 합

$\hat{H}_0 + \underbrace{\hat{H}_{\mathrm{rel}} + \hat{H}_{\mathrm{SO}}}$

미세 구조

- 항 기호

섭동론